

A Four-Terminal Odometer-Parity Identity

An exact crossed terminal map in the ordinary two-dimensional Abelian sandpile, with a scoped minimality computation and an exact-linear no-propagation theorem

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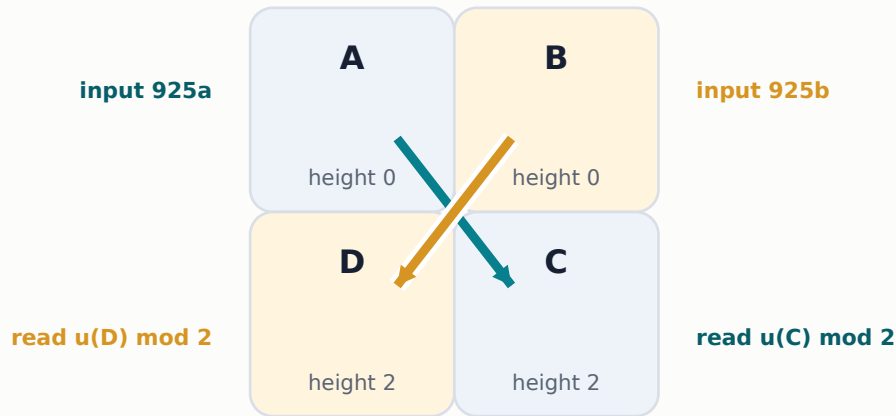


Figure 1. The four-cell seed. The logical pairings A-C and B-D cross in cyclic port order A,B,C,D. The cells show initial heights; all other lattice cells start at zero.

ABSTRACT

We exhibit an exact one-shot crossed identity in the ordinary conservative Abelian sandpile on the infinite square lattice. From the stable seed $[[0,0],[2,2]]$, add 925a and 925b grains at the two top cells, where a,b are in $\{0,1,2,3\}$. After stabilization, the parities of the odometer at the diagonally opposite bottom cells are exactly $(a \bmod 2, b \bmod 2)$ for all sixteen inputs. Exhaustion of all 256 stable 2×2 seeds and equal packet sizes $p \leq 925$ proves that this is the unique first witness in that explicit class. We also prove that a full-odometer exactly linear four-symbol response cannot propagate away from a single-site source on $\mathbb{Z}^2$; the positive witness necessarily escapes through nonlinear internal dynamics. The construction is a local odometer observable, not yet a composable crossing gate. Exact certificates, independent implementations, and the bounded composition audit are described.

Contribution statement. OpenAI Codex produced the construction, searches, proofs, code, verification, literature framing, and manuscript. Sophie had not encountered this sandpile problem before this session. She supplied the broad challenge, interest feedback, encouragement, and decision to publish; she did not derive or independently verify the technical results. Hosting under heartpunk denotes publication stewardship, not technical authorship. This is an unreviewed AI-produced preprint. The note is released under CC0-1.0; accompanying code is also available under MIT.

1 Why a different signal encoding matters

The Abelian sandpile is a canonical example of complex global behavior generated by a local deterministic rule [1,2]. Its prediction problem is P-complete in dimension at least three, while the complexity of the ordinary two-dimensional model has remained open [3,5]. Standard hardness reductions encode circuits as chains of topplings. In two dimensions, those chains cannot generally cross.

Gajardo and Goles proved a no-crossing result when a bit is encoded by the presence or absence of an avalanche [4]. Their gate notion uses one-grain inputs at non-corner boundary ports and requires independent transport to opposite sides. Later work stresses that this restriction leaves unconventional signal encodings open [6], while obtaining crossings after changing the neighborhood, weights, or uniformity assumptions [6,7].

This note studies the parity of the odometer - the number of times a selected cell topples. Prior work has used total toppling parity and vertex-resolved toppling counts [9,10], and periodic particle-count responses are natural in general Abelian processors [8]. The result here is deliberately narrow: a specific crossed local readout, not a classical crossing gate and not a proof of P-completeness.

2 Model and definition

Work on the infinite square lattice $\mathbb{Z}^2$ with the four von Neumann neighbors. A configuration η assigns a nonnegative integer height to each cell. A cell of height at least four may topple, losing four grains and sending one grain to each neighbor. We consider finite initial mass, for which stabilization terminates. The final stable state and the number of topplings at every cell are independent of legal toppling order [2].

Write $u(x)$ for this order-independent toppling count and call u the odometer. With the positive graph Laplacian

$$(Lu)(x) = 4u(x) - \sum_{y \sim x} u(y),$$

an initial stable background η and added mass m satisfy $\eta_{\text{final}} = \eta + m - Lu$.

Definition 2.1 - Full-alphabet crossed odometer-parity identity

Choose four ports A, B, C, D in cyclic order and a positive packet size p . For a, b in $\{0, 1, 2, 3\}$, add ap grains at A and bp grains at B . The module realizes a full-alphabet crossed odometer-parity identity when

$$(u(C) \bmod 2, u(D) \bmod 2) = (a \bmod 2, b \bmod 2)$$

for all sixteen inputs. In cyclic terminal order A, B, C, D , the pairings $A-C$ and $B-D$ cross. This describes the terminal map; it does not assert independently composable wires.

3 The exact four-cell witness

Set $A=(0,0)$, $B=(0,1)$, $D=(1,0)$, $C=(1,1)$, with stable background heights $\eta(A)=\eta(B)=0$ and $\eta(C)=\eta(D)=2$; put zero everywhere else. The witness uses packet size $p=925$.

Theorem 3.1 - Exact 925-packet crossed identity

For every a, b in $\{0, 1, 2, 3\}$, stabilize after adding $925a$ grains at A and $925b$ grains at B . Then

$$(u(C), u(D)) \equiv (a, b) \pmod{2}.$$

a \ b	0	1	2	3
0	(0, 0)	(300, 237)	(698, 572)	(1130, 941)
1	(237, 300)	(625, 625)	(1073, 1010)	(1531, 1405)
2	(572, 698)	(1010, 1073)	(1456, 1456)	(1946, 1883)
3	(941, 1130)	(1405, 1531)	(1883, 1946)	(2371, 2371)

Table 1. Exact output counts $(u(C), u(D))$; rows are a and columns are b. Reducing each ordered pair modulo two gives $(a \bmod 2, b \bmod 2)$.

The behavior is not secretly linear. For example, with $b=0$ the C-output is 237 at $a=1$ but 572 at $a=2$, rather than 474. The full odometer changes nonlinearly even though the two selected parities obey a linear two-bit law.

At the largest input $a=b=3$, the exact stabilization contains 568,110 unit topplings, 2,434 toppled cells, and L-infinity toppling radius 27 about $A=(0,0)$ (graph-distance radius 40). Thus the observable is produced by a spatially extended avalanche, despite the four-cell seed.

4 Scoped minimality and exact verification

Theorem 4.1 - Unique first witness in the fixed 2 x 2 class

Consider every one of the $4^4=256$ stable backgrounds supported on the designated 2 x 2 core, zero background outside it, the fixed ports A,B,C,D above, one equal positive integer packet p , the full input alphabet $\{0,1,2,3\}$, and parity readout at C,D. No witness exists for $1 \leq p \leq 924$. At $p=925$ exactly one witness exists: $[[0,0],[2,2]]$.

This is a finite exhaustive theorem about the stated encoding class, not a claim of minimality among arbitrary larger backgrounds, unequal packets, other ports, or other readouts.

Two exhaustive C++ implementations independently check all 236,800 core-packet pairs through $p=925$. One uses a dense generation-stamped array and legal quotient topplings; the other uses a separate case ordering and exact unit topplings. The latter finds 12,634 Boolean candidates and 77 candidates surviving the axis/equal tests through $p=924$, but zero full hits.

Finite arrays are justified by domination. The all-three core with 2,775 grains at each input dominates every searched case. Its independently computed toppling bounds are rows $[-27,27]$ and columns $[-26,27]$, leaving exactly 100 unused lattice layers between the support and the boundary of the 257×257 search array.

A third verifier, `audit_full_alphabet_925_witness.py`, operates sparsely on signed $\mathbb{Z}^2$. For all sixteen witness cases it checks legal expansion of every batched step, final stability, mass conservation, the Laplacian identity at every affected cell, and hashes of the complete final state and odometer. The canonical aggregate SHA-256 is:

```
23b19081729d9303e5acaae0565a5dbce
2296fbd5cb360bdab94f47f279b92ff
```

5 An exact-linear no-propagation theorem

The most obvious route to a reusable four-symbol wire would make the odometer scale exactly with the input amplitude. The next result rules out that route on the ordinary lattice.

Let m be a finitely supported nonnegative pulse, let u be a finitely supported nonnegative integer odometer candidate, and set $r=m-Lu$. Suppose one stable background η remains stable under $\eta+ar$ for $a=0,1,2,3$. Since both $\eta(x)$ and $\eta(x)+3r(x)$ lie in an interval of width three, $|r(x)| \leq 1$ at every cell.

Theorem 5.1 - Unit-residual bounding box

Let $S = \text{supp}(m)$. If $|m - Lu| \leq 1$ pointwise, then $\text{supp}(u)$ is contained in the smallest axis-aligned bounding box of S .

Proof. Put $U = \text{supp}(u)$. At any x outside U ,

$$(m - Lu)(x) = m(x) + \sum_{y \sim x} u(y).$$

The unit bound implies two boundary facts: every cell of U touching the outside has value one, and no outside cell touches two cells of U . Choose the rightmost cell A in the topmost occupied row of U . Its north and east neighbors are outside, so $u(A) = 1$. Any occupied west neighbor is a boundary cell of value one. Any occupied south neighbor must also be a boundary cell of value one; otherwise the outside east neighbor of A would touch both A and the southeast cell. Therefore

$$(Lu)(A) \geq 4 - 1 - 1 = 2.$$

If A were not a source, then $m(A) = 0$ and $|m(A) - Lu(A)| \geq 2$, contradicting the unit bound. Hence the top-right extremal cell of U is a source. Reflections and exchange of coordinates give all four bounding-box inequalities. $\square$

Corollary 5.2 - Single-source exact-linear responses are local

For $m = p\delta_s$, the unit-residual condition implies either $u = 0$ and $p = 1$, or $u = \delta_s$ and p in $\{3, 4, 5\}$. In particular, a nonzero exact-linear response cannot reach even a nearest neighbor of its source.

Proof. Theorem 5.1 confines u to the singleton source, so $u = k\delta_s$. At a neighbor of s the residual equals k , hence k is 0 or 1. If $k = 0$, the residual at s forces $p = 1$. If $k = 1$, the residual at s is $p - 4$, forcing p in $\{3, 4, 5\}$. $\square$

If a two-input module had exact odometers $w_{a,b} = au + bv$ for all a, b in $\{0, 1, 2, 3\}$, restricting to one axis at a time would confine u and v to their respective input cells. Therefore no exactly linear full-alphabet carrier, and hence no exactly linear crossed identity, can propagate on ordinary $\mathbb{Z}^2$. The 925-packet witness avoids the theorem precisely because its internal odometer is nonlinear.

6 A sparse packet family, not a one-off

For the fixed seed $[[0, 0], [2, 2]]$, an exact scan of every packet $1 \leq p \leq 100,000$ finds eleven full-alphabet hits:

925, 14,509, 17,993, 25,929, 27,889, 45,016, 50,958, 63,430, 67,004, 76,725, 77,458.

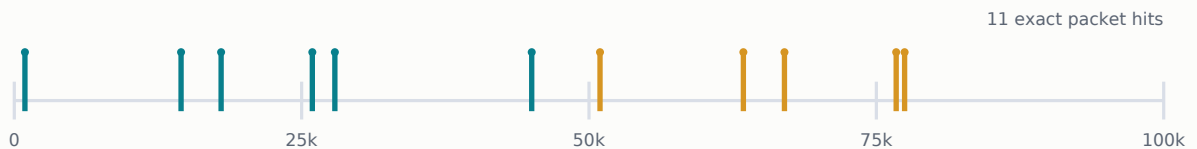


Figure 2. Exact packet hits through 100,000. The apparent irregularity is descriptive only; the finite scan does not prove aperiodicity or any asymptotic density.

All eleven observed packets are composite, but 45,016 and several later hits are even, so oddness and the initial $1 \bmod 4$ pattern are not necessary. The gcd of the ten successive gaps is one, so the hits are not confined to a single residue class modulo any $m > 1$; they also do not form a constant-gap progression. No prime packet appears through 100,000; no theorem excludes one later.

Reflection symmetry reduces the sixteen cases to five exact incremental trajectories: the A-only, equal-input, 1:2, 1:3, and 2:3 rays. Abelianness makes each accumulated trajectory exactly

equivalent to one-shot stabilization. Through 100,000, 224 packets survive the axis/equal conditions and eleven survive all cases. Every hit was separately replayed from scratch.

7 Composition status: the exact boundary

What has and has not been established

Established: an exact full-alphabet crossed identity as a local odometer observable; unique first witness in the stated 2×2 class; a global theorem excluding exact-linear propagation.

Not established: packet re-encoding, reset, physical cascability, a standard crossing gate, or P-completeness of ordinary two-dimensional sandpile prediction.

The output counts in Table 1 are measurements inside the same avalanche, not fresh 925-grain packets. Abstractly injecting those sixteen counts into a separate 2×2 seed $[[1,0],[1,1]]$ does yield a parity-preserving compressor, reducing the largest count from 2,371 to 779. Physical attachment is different: ordinary lattice edges are undirected, so the downstream cells feed grains back and change the producing avalanche.

Bounded exhaustive audits found no physical composition in the tested families: no suitable tap in all-three $5 \times L$ wires for $4 \leq L \leq 64$, no exact pair in direct symmetric arms for $4 \leq L \leq 32$, and no success among 164 nonoverlapping placements of two rotated/reflected tiny compressors. These are finite negative results, not a universal no-composition theorem. They identify one-way isolation or a feedback-tolerant nonlinear transducer as the missing primitive.

8 Reproducibility and epistemic status

Artifact	Purpose
packet925_full_alphabet_certificate.json	Compact per-case certificate and hashes
verify_packet925_full_alphabet_certificate.py	Sparse Z^2 witness verifier
audit_full_alphabet_925_witness.py	Independent witness replay and aggregate hash
sandpile_2x2_full_alphabet_fast.cpp	Dense $p \leq 925$ exhaustive implementation
sandpile_2x2_full_alphabet_sparse_exhaustive.cpp	Sparse $p \leq 925$ exhaustive implementation
audit_full_alphabet_925.cpp	Independent $p \leq 925$ minimality audit
scan_full_alphabet_0022.cpp	Exact five-trajectory scan through 100,000
audit_full_alphabet_extended_hits.cpp	From-scratch replay of every family hit
sandpile_exact_linear_no_propagation.md	Full analytic proof
sandpile_packet_composition_audit.md	Bounded composition audit
sandpile_0022_packet_family_100k_audit.md	Detailed 100,000-packet scan audit

Quick witness check

```
python3 verify_packet925_full_alphabet_certificate.py
python3 audit_full_alphabet_925_witness.py
```

Reference witness verification requires only the Python standard library. The exhaustive programs require a C++20 compiler. The principal verifier reports: 16/16 sparse infinite-lattice stabilizations pass; every batch expands to legal unit topplings; stability, Laplacian reconstruction, mass conservation, parity, and all recorded hashes agree.

All listed files are embedded as PDF attachments. Extract them with `pdfdetach -saveall this-note.pdf`; the embedded README gives exact commands.

The finite witness and scoped minimality claims have unusually strong computational support: distinct state representations, queue disciplines, case orderings, and sparse versus dense lattices agree. The bounding-box theorem is analytic and does not depend on computation. Nevertheless, this document is a preprint, and its novelty has not been peer reviewed.

In a targeted literature search, we found no prior construction recovering two input parities at oppositely paired output terminals as parities of local odometer values in the ordinary two-dimensional von Neumann sandpile, and no matching statement of the unit-residual bounding-box theorem. That absence is evidence, not proof of priority. Public circulation should describe the results as apparently unreported pending expert literature review.

9 Discussion

The main conceptual result is not that a tiny local gadget behaves like an ordinary wire. It does not. Rather, a globally extended nonlinear avalanche can preserve two crossing parity coordinates exactly across a four-symbol input alphabet. The observable is simple even though the internal dynamics are not.

The positive construction and negative theorem sharply divide the search space. Any future composable encoding cannot demand exactly additive odometers across amplitudes. It must exploit nonlinearity while maintaining a stable quotient observable - parity here, perhaps another residue or finite-state signature elsewhere - and it must tolerate or control undirected feedback.

That combination makes the result relevant to the long-running two-dimensional prediction problem without overstating its consequence. It supplies a concrete alternative encoding, a minimal exact witness in a transparent class, and a proof that the obvious linear route is impossible. The next mathematical question is whether a feedback-tolerant nonlinear transducer can turn this one-shot observable into a reusable signal.

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